

CHM2045C: Final Exam Comprehensive Study Guide

Chemistry Study Guide Draft

Spring 2026

Chapter 1: Foundations: The Chemical Toolbox

This chapter establishes the fundamental quantitative and qualitative tools required for chemical analysis, focusing on measurement, classification, and naming.

Density and Volume Displacement

Density is an intrinsic property relating mass and volume. For irregular solids, volume is determined by the displacement of a liquid.

Key Formula (Must Memorize)

$$d = \frac{m}{V} \quad \text{and} \quad V_{\text{object}} = V_{\text{final}} - V_{\text{initial}}$$

1. Identify given mass (g) and volume (mL or cm³).
2. Use displacement to find V if necessary (Archimedes' Principle).
3. Solve for the unknown variable using algebraic rearrangement.

Chemical Nomenclature

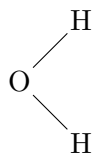
- **Ionic Compounds:** Name the cation followed by the anion. Use Roman numerals for transition metals with variable oxidation states (e.g., FeCl₃ is iron(III) chloride).
- **Acids:** Binary acids (H + element) use the prefix *hydro-* and suffix *-ic* (e.g., HCl(aq) is hydrochloric acid). Oxyacids use *-ic* for *-ate* ions and *-ous* for *-ite* ions.
- **Hydrates:** Append the number of water molecules using Greek prefixes (e.g., CuSO₄ · 5 H₂O is copper(II) sulfate pentahydrate).

Classification of Matter

Matter is classified by its composition and uniformity at the particulate level.

- **Elements:** Pure substances consisting of one type of atom (e.g., O₂).
- **Compounds:** Pure substances with fixed ratios of different atoms (e.g., H₂O).
- **Mixtures:** Physical blends of substances. **Homogeneous** (uniform) or **Heterogeneous** (non-uniform).

Particulate Representation of a Water Molecule (H_2O):



Metric System, Units, and Temperature

Mastery of prefixes is essential for dimensional analysis and unit comparison.

Prefix	Nano (n)	Micro (μ)	Milli (m)	Kilo (k)
Factor	10^{-9}	10^{-6}	10^{-3}	10^3

Temperature Conversion (Provided)

$$T_K = T_{\circ C} + 273.15$$

Note: Kelvin is mandatory for all gas law and thermodynamic calculations. Absolute zero is 0 K.

Chapter 2: Atomic Structure

Understanding the atom involves identifying its constituent subatomic particles and their contribution to the element's identity and mass.

Nuclide Notation and Subatomic Particles

The identity of an atom is defined by its atomic number (Z), while its mass is determined by the sum of protons and neutrons.

Key Formula (Must Memorize)

$$A = Z + N \quad \text{and} \quad \text{Charge} = \text{Protons} - \text{Electrons}$$

In the symbol ${}_Z^AX^{\text{charge}}$:

- **Protons:** Equal to Z (Atomic Number). Defines the element.
- **Neutrons:** Calculated as $A - Z$ (Mass Number - Atomic Number).
- **Electrons:** Calculated as $Z - (\text{Charge})$. In neutral atoms, $e^- = p^+$.

Average Atomic Mass

The mass reported on the periodic table is a weighted average of all naturally occurring isotopes based on their abundance.

Key Formula (Must Memorize)

$$\text{Avg. Atomic Mass} = \sum_i (\text{Mass of Isotope}_i \times \text{Fractional Abundance}_i)$$

Technique: Convert percentages to decimals (e.g., 75% $\rightarrow$ 0.75) to find the "fractional abundance" before multiplying by isotopic mass.

Isotopes vs. Ions

- **Isotopes:** Atoms of the same element (same Z) with different numbers of neutrons (N), resulting in different mass numbers (A).
- **Ions:** Atoms that have gained or lost electrons, resulting in a net positive (cation) or negative (anion) charge.

Chapter 3: Electronic Structure and Periodic Trends

Electromagnetic Radiation and the Bohr Model

Light behaves as both a wave and a particle (photon). The energy of an electron in an atom is quantized.

Light and Photon Energy (Must Memorize)

$$c = \lambda\nu \quad \text{and} \quad E = h\nu = \frac{hc}{\lambda}$$
$$\Delta E = -2.18 \times 10^{-18} \text{J} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

Note: h and c are usually provided, but the relationships between E , λ , and ν must be mastered.

Quantum Numbers

Four quantum numbers describe the unique quantum state (or "address") of an electron:

1. **Principal (n):** Energy level (1, 2, 3...).
2. **Angular Momentum (l):** Orbital shape (0 to $n - 1$). $s = 0, p = 1, d = 2, f = 3$.
3. **Magnetic (m_l):** Orbital orientation ($-l$ to $+l$).
4. **Spin (m_s):** Direction of electron spin ($+1/2$ or $-1/2$).

Electron Configurations

Follow three core principles to populate orbitals in the ground state:

- **Aufbau Principle:** Electrons fill the lowest energy orbitals first (1s, 2s, 2p...).
- **Pauli Exclusion Principle:** No two electrons can have the same four quantum numbers (max 2 e^- per orbital with opposite spins).
- **Hund's Rule:** Electrons fill degenerate orbitals singly with parallel spins before pairing.

Noble Gas Shorthand: Use the preceding noble gas to represent core electrons (e.g., Sn is [Kr] 5s² 4d¹⁰ 5p²).

Periodic Trends: Ionization Energy (IE)

Ionization Energy is the energy required to remove an electron from a gaseous atom.

- **Across a Period:** IE generally increases due to increasing effective nuclear charge (Z_{eff}).
- **Down a Group:** IE generally decreases due to increasing atomic radius and electronic shielding.

Chapter 4: Formula Calculations and the Mole

Core Concepts

The mole is the fundamental unit in chemistry for measuring the amount of a substance. It provides a bridge between the microscopic world of atoms and molecules and the macroscopic world of grams and liters.

- **Avogadro's Number:** 1 mole = 6.022×10^{23} representative particles.
- **Molar Mass:** The mass in grams of one mole of a substance (g mol^{-1}).
- **Percent Composition:** The percentage by mass of each element in a compound.

Avogadro Constant (Provided)

$$N_A = 6.022 \times 10^{23} \text{ particles mol}^{-1}$$

Mole Calculations (Must Memorize)

$$n = \frac{m}{MM} \quad \text{and} \quad N = n \times N_A$$

Problem-Solving Skills

1. Mass-to-Particle Conversions:

$$\bullet \text{ Mass (g)} \xrightarrow{\div MM} \text{Moles} \xrightarrow{\times N_A} \text{Particles}$$

2. Calculating Empirical Formulas:

- a) Assume a 100 g sample (convert % directly to grams).
- b) Convert grams of each element to moles using atomic masses.
- c) Divide all mole values by the smallest mole value obtained.
- d) If necessary, multiply by a small integer to reach whole numbers.

3. Percent Mass Composition:

Key Formula (Must Memorize)

$$\text{Mass \% of element} = \left(\frac{\text{mass of element in 1 mol compound}}{\text{molar mass of compound}} \right) \times 100\%$$

Chapter 5: Stoichiometry

Core Concepts

Stoichiometry involves using the relationships between reactants and products in a balanced chemical equation.

Reaction Efficiency (Must Memorize)

$$\text{Percent Yield} = \left(\frac{\text{Actual Yield}}{\text{Theoretical Yield}} \right) \times 100\%$$

Problem-Solving Skills

1. Mass-to-Mass Stoichiometry:



2. Identifying the Limiting Reactant:

- Convert the given amounts of all reactants to moles.
- Divide the moles of each reactant by its coefficient in the balanced equation.
- The reactant with the smallest resulting value is the **limiting reactant**.

Chapter 6: Solutions and Aqueous Reactions

Core Concepts

Concentration and Dilution (Must Memorize)

$$M = \frac{n}{V} \quad \text{and} \quad M_1V_1 = M_2V_2$$

Solubility Reference

Table 1: Solubility Guidelines for Ionic Compounds (Must Memorize)

Always Soluble	NO_3^- , $\text{C}_2\text{H}_3\text{O}_2^-$, Group 1 (Li^+ , Na^+ , K^+), NH_4^+
Generally Soluble	Cl^- , Br^- , I^- (except Ag^+ , Pb^{2+} , Hg_2^{2+}) SO_4^{2-} (except Ba^{2+} , Sr^{2+} , Ca^{2+} , Pb^{2+} , Ag^+)
Generally Insoluble	S^{2-} , OH^- (except Group 1, NH_4^+ ; Ca^{2+} , Sr^{2+} , Ba^{2+} are soluble) CO_3^{2-} , PO_4^{3-} (except Group 1 and NH_4^+)

Oxidation States

- F = -1; O = -2 (usually); H = +1 (with non-metals).
- The sum of oxidation states must equal the overall charge.

Chapter 7: Heat and Enthalpy

Calorimetry

Specific Heat Equation (Provided)

$$q = mc_s\Delta T$$

Reaction Enthalpy Calculations

Enthalpy Calculations (Must Memorize)

From Heats of Formation:

$$\Delta H_{\text{rxn}}^{\circ} = \sum n\Delta H_f^{\circ}(\text{products}) - \sum m\Delta H_f^{\circ}(\text{reactants})$$

From Bond Energies:

$$\Delta H_{\text{rxn}}^{\circ} = \sum (\text{bonds broken}) - \sum (\text{bonds formed})$$

Crucial Distinction: Formation uses (Products - Reactants), while Bond Energy uses (Reactants/Broken - Products/Formed).

Chapter 8: Structure and Bonding

Lattice Energy

Lattice energy increases with increasing ion charge and decreasing ion size.

Coulomb's Law (Must Memorize)

$$E_{\text{lattice}} \propto \frac{Q_1 Q_2}{d}$$

Formal Charge

Formal Charge (Must Memorize)

$$FC = (\text{Valence } e^-) - (\text{Non-bonding } e^-) - \frac{1}{2}(\text{Bonding } e^-)$$

Chapter 9: States of Matter

Gas Laws and Constants

Gas Laws and Constants (Provided)

Constants:

$$R = 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1}$$

$$1 \text{ atm} = 760 \text{ mmHg} = 760 \text{ Torr}$$

Laws:

$$PV = nRT \quad \text{and} \quad \frac{P_1 V_1}{n_1 T_1} = \frac{P_2 V_2}{n_2 T_2}$$

Graham's Law of Effusion:

$$\frac{\text{rate}_1}{\text{rate}_2} = \sqrt{\frac{M_2}{M_1}}$$

Gas Density

Gas Density (Must Memorize)

$$d = \frac{PM}{RT}$$

Addendum: Exam Formula & Constant Reference

Provided Information

- **Temperature:** $T_K = T_{\circ C} + 273.15$
- **Avogadro Constant:** $N_A = 6.022 \times 10^{23} \text{mol}^{-1}$
- **Gas Constant:** $R = 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1}$
- **Pressure:** $1 \text{ atm} = 760 \text{ mmHg} = 760 \text{ Torr}$
- **Ideal Gas Law:** $PV = nRT$
- **Combined Gas Law:** $\frac{P_1 V_1}{n_1 T_1} = \frac{P_2 V_2}{n_2 T_2}$
- **Graham's Law:** $\frac{\text{rate}_1}{\text{rate}_2} = \sqrt{\frac{M_2}{M_1}}$
- **Specific Heat:** $q = mc_s \Delta T$

Information to Memorize

- **Atomic Mass:** $\sum(\text{mass} \times \text{fraction abundance})$
- **Light:** $c = \lambda \nu$ and $E = h\nu = hc/\lambda$
- **Rydberg:** $\Delta E = -2.18 \times 10^{-18} \text{J} (1/n_f^2 - 1/n_i^2)$
- **Molarity:** $M = n/V$
- **Enthalpy (Formation):** $\sum \Delta H_f^\circ(\text{prod}) - \sum \Delta H_f^\circ(\text{react})$
- **Enthalpy (Bonds):** $\sum(\text{broken}) - \sum(\text{formed})$
- **Lattice Energy:** $E \propto Q_1 Q_2 / d$
- **Solubility:** (See Table in Chapter 6)